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*Technical Report*

## 3rd Generation Partnership Project;
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# Contents

# Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

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Version x.y.z

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x the first digit:

1 presented to TSG for information;

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y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

# Introduction

Work has started in ITU and 3GPP to develop requirements and specifications for new radio (NR) systems, as in the Recommendation ITU-R M.2083 "Framework and overall objectives of the future development of IMT for 2020 and beyond", as well as 3GPP SA1 study item New Services and Markets Technology Enablers (SMARTER) and SA2 study item Architecture for NR System. 3GPP has to identify and develop the technology components needed for successfully standardizing the NR system timely satisfying both the urgent market needs, and the more long-term requirements set forth by the ITU-R IMT-2020 process. In order to achieve this, evolutions of the radio interface as well as radio network architecture are considered in the study item "New Radio Access Technology" [1].

# 1 Scope

The present document covers the Radio Access Architecture and Interface aspects of the study item “New Radio Access Technology” [1]. The purpose of this TR is to record the discussion and agreements that arise in the specification of the “New Radio Access Technology” from an Access Architecture and Interface specification point of view.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.


[1] 3GPP RP-160671: "New SID Proposal: Study on New Radio Access Technology".

[2] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[3] 3GPP TS 36.401: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Architecture description".

[4] 3GPP TR 22.862: "FS_SMARTER - Critical Communications".

[5] 3GPP TR 38.913: "Study on Scenarios and Requirements for Next Generation Access Technologies".

[6] 3GPP TR 23.799: "Study on Architecture for Next Generation System".

[7] RP-161266, "5G Architecture Options- Full Set". Joint RAN/SA Meeting, Busan, S. Korea, June 2016.

[8] R3-161646, "Migration path towards NR with EPC", NTT DOCOMO, INC.

[9] R3-161772, "AT&T Views on 5G Architecture Evolution: Early Phase 1 To Mature Phase 2 Deployment", AT&T.

[10] R3-161809, "Analysis of migration paths towards RAN for new RAT", CMCC.

[11] R3-161813, "Transport requirement for CU&DU functional splits options", CMCC.

[12] 3GPP TS 36.300: "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Overall description; Stage 2".

[13] 3GPP TS 36.423: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 application protocol (X2AP)".

# 3 Definitions and abbreviations

## 3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [2] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [2].

**example:** text used to clarify abstract rules by applying them literally.

**eLTE eNB:** The eLTE eNB is the evolution of eNB that supports connectivity to EPC and NG Core.

**Evolved E-UTRA:** An evolution of E-UTRA for operation in the NextGen system (see TR 23.799 [6]).

**New RAN:** A Radio Access Network which supports either NR or an evolved E-UTRA or both, interfacing with the next generation core (see TR 23.799 [6]).

**New Radio:** A new radio access technology which is studied under [1].

**Network Slice:** A Network Slice is a network created by the operator customized to provide an optimized solution for a specific market scenario which demands specific requirements with end to end scope as described in TR 23.799 [6].

**Non-standalone NR**: A deployment configuration where the gNB requires an LTE eNB as anchor for control plane connectivity to EPC, or an eLTE eNB as anchor for control plane connectivity to NGC.

**Non-standalone Evolved E-UTRA**: A deployment configuration where the eLTE eNB requires a gNB as anchor for control plane connectivity to NGC.

## 3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [2] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [2].

| | |
|---|---|
| NGC | Next Generation Core |
| NR | New Radio |

# 4 Objectives and requirements

Editor's note: Intention is to capture objectives of RAN3 study in reference to requirements in TR 38.913 and objectives identified in the NR technology SID in RP-160671. The intended the use of this TR and principles for accepting texts to this TR may also be captured if needed.

## 4.1 Support for high reliability low latency services

The NR shall be able to support the highly reliable (i.e. with low ratio of erroneous packets), and low latency services.

Support for high reliability services is subject to the following requirements and assumptions:

- High Reliability:

  High reliability is about providing high likelihood of delivering error free packets through the 3GPP system within a bounded latency. A performance metric for high reliability is the ratio of successfully delivered error free packets within a delay bound over the total number of packets. The required ratio and latency bound may be different for different URLLC use cases.

- High Availability:

  High availability is related to a communication path through the 3GPP system providing reliable services. This communication path between the communication end points is made up of radio links as well as transport links and different HW and SW functions. The NR should provide high availability, in addition to deploy redundant components and links for these radio, transport and HW/SW.

- Low Latency:

According to TR 38.913, the NR should support latencies down to 0.5 ms UL/DL for URLLC.

It is FFS whether a new function is needed at the NR to support high reliability low latency services.

It is FFS whether and how end to end delays will be considered.

# 5 Deployment scenarios

## 5.1 General

The following example scenarios should be considered for support by the New RAN architecture.

Although it is not always explicitly specified, it should be assumed that an inter BS interface may be supported between an NR BS and other NR BSs or LTE eNBs.

The Heterogeneous deployment comprises in the same geographical area two or more deployments as defined in sections 5.1 to 5.5 of the present document.

## 5.2 Non-centralised deployment

In this scenario, the full protocol stack is supported at the NR base station (NR BS) e.g. in a macro deployment or indoor hotspot environment (could be public or enterprise). The NR BS can be connected to "any" transport. It is assumed that the NR BS is able to connect to other NR or LTE BSs via a RAN interface.

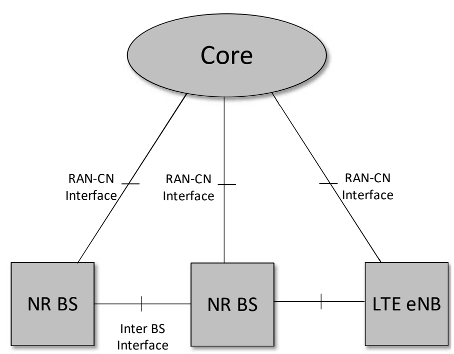



**Figure 5.2-1: Non-centralised deployment**

## 5.3 Co-sited deployment with LTE

In this scenario the NR functionality is co-sited with LTE functionality either as part of the same base station or as multiple base stations at the same site. Co-sited deployment can be applicable in all NR deployment scenarios e.g. Urban Macro. In this scenario it is desirable to fully utilise all spectrum resources assigned to both RATs by means of load balancing or connectivity via multiple RATs (e.g. utilising lower frequencies as coverage layer for users on cell edge).

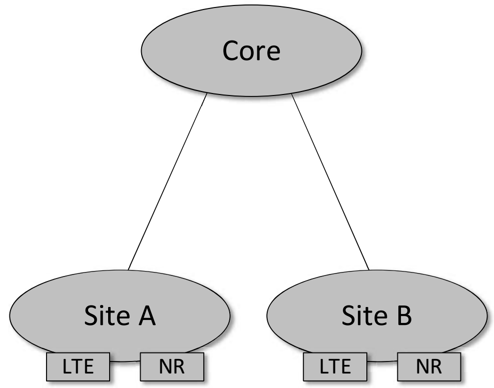



**Figure 5.3-1: Co-sited deployment with LTE**

## 5.4 Centralized deployment

NR should support centralization of the upper layers of the NR radio stacks.

Different protocol split options between Central Unit and lower layers of NR BS nodes may be possible. The functional split between the Central Unit and lower layers of NR BS nodes may depend on the transport layer.

High performance transport between the Central Unit and lower layers of NR BS nodes, e.g. optical networks, can enable advanced CoMP schemes and scheduling optimization, which could be useful in high capacity scenarios, or scenarios where cross cell coordination is beneficial.

Low performance transport between the Central Unit and lower layers of NR BS nodes can enable the higher protocol layers of the NR radio stacks to be supported in the Central Unit, since the higher protocol layers have lower performance requirements on the transport layer in terms of bandwidth, delay, synchronization and jitter.

Both non co-sited deployment and co-sited deployment with LTE can be considered for this scenario.

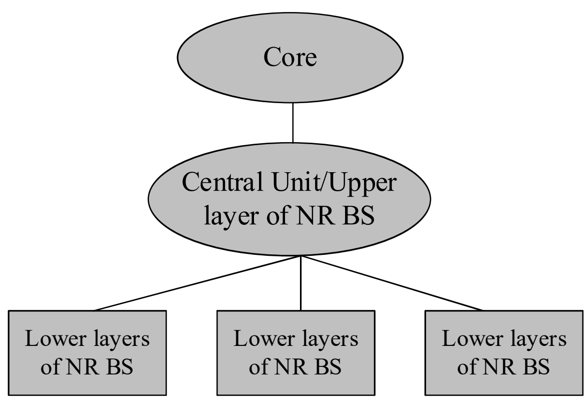



**Figure 5.4-1: Centralized deployment**

## 5.5 Shared RAN deployment

Similar to LTE, NR should support shared RAN deployments.

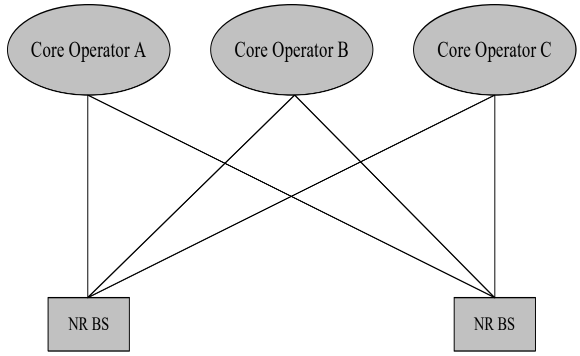



**Figure 5.5-1: Shared RAN deployment**

# 6 Radio access network architecture for New RAT

## 6.1 Functional split

Editor's note: Intention is to capture high level list of functions to be allocated between RAN and CN and within the RAN.

### 6.1.1 RAN-CN functional split

#### 6.1.1.1 Functions for New RAN

NOTE 1: The further detail of the functionalities needs further study.

**Functions similar to E-UTRAN as listed in TS 36.401 [3]**

- Transfer of user data
- Radio channel ciphering and deciphering
- Integrity protection
- Header compression
- Mobility control functions:
    - Handover
- Inter-cell interference coordination
- Connection setup and release
- Load balancing
- Distribution function for NAS messages
- NAS node selection function
- Synchronization

- Radio access network sharing
- Paging
- Positioning

**Functions specific for New RAN:**

- Network Slice support
  - This function provides the capability for NR RAN to support network slicing
- Tight Interworking with LTE
  - This function enables tight interworking between NR and evolved E-UTRA by means of data flow aggregation. This function includes at least dual connectivity. Interworking with evolved E-UTRA is supported for collocated and non-collocated site deployments.
- Multi-connectivity
  - This function provides means for connectivity between an New RAN node and multiple New RAN nodes by means of data flow aggregation.
- evolved E-UTRA-NR handover through a New RAN interface
  - This function provides means for evolved E-UTRA-NR handover via the direct interface between an eLTE eNB and a NR BS.

Editor's Note: How to cover options 3/3a is FFS.

- Session Management
  - This function provides means for the NG-Core to create/modify/release a context in the New RAN associated with a particular PDU session of a UE.

**New RAN may also include other functions:**

- Contacting UEs in inactive mode (feasibility of UE Inactive mode to be further studied and coordinated with RAN2)
  - In the inactive mode, the UE context is stored in RAN and UP data is buffered in RAN.
- Direct services support (further study related with D2D, coordinate with RAN1, RAN2)
  - This function provides communication whereby UEs can communicate with each other directly
- Interworking with non-3GPP systems
  - This function provides interworking between NR and Non-3GPP RAT (e.g. WLAN).
- evolved E-UTRA/E-UTRA - NR handover via CN
  - This function provides means for evolved E-UTRA/E-UTRA - NR handover via CN.

  Note: Support of the E-UTRA - NR HO via CN function depends on progress in SA2. When discussing solutions to support this function, RAN3 needs to consider factors such as adaptation of the source RAN/CN to the target RAN/CN

- evolved E-UTRAN-NR inactive mode mobility using a direct interface between an eLTE eNB and an NR BS.
  - This function provide means for UE mobility in a RAN controlled inactive mode between NR and evolved E-UTRAN using a direct interface between an eLTE eNB and an NR BS for context fetching and paging.

Editor's Note: This approach is pending discussions on "light connection" and respective discussions on paging.

#### 6.1.1.1.1 Support for a UE operational mode during periods of no traffic [FFS RAN2]

The RAN3 impact of a UE operational mode during periods of no traffic should be explored in the Next Generation Systems and New Radio (NR) work. For instance upon reception of DL data, New RAN could notify the UE.

## 6.1.2 Functional split between central and distributed unit

### 6.1.2.1 General description of split options

In the study item for a new radio access technology, 3GPP is expected to study different functional splits between central and distributed units. E-UTRA protocol stack is taken as a basis for further discussion, with the understanding that the conclusions may need to be revisited, once RAN2 defines the protocol stack for NR. The following functional splits between central and distributed unit are possible, as illustrated in Figure 6.1.2.1-1.

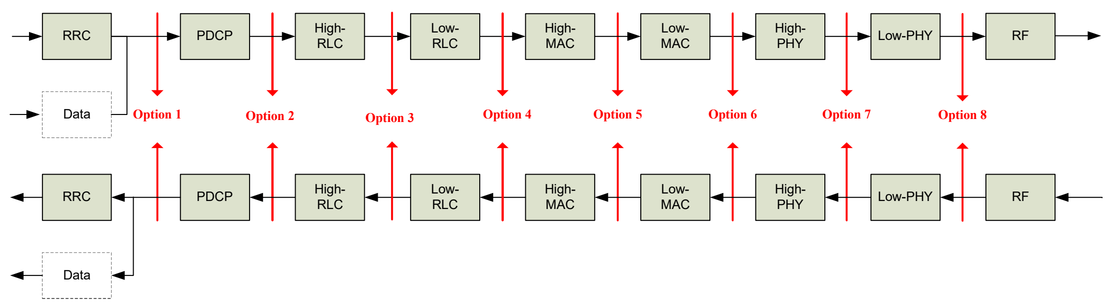



**Figure 6.1.2.1-1: Function Split between central and distributed unit**

**Option 1 (1A-like split)**

- The function split in this option is similar as 1A architecture in DC. RRC is in the central unit. PDCP, RLC, MAC, physical layer and RF are in the distributed unit.

**Option 2 (3C-like split)**

- The function split in this option is similar as 3C architecture in DC. RRC, PDCP are in the central unit. RLC, MAC, physical layer and RF are in the distributed unit.

**Option 3 (intra RLC split)**

- Low RLC (partial function of RLC), MAC, physical layer and RF are in distributed unit. PDCP and high RLC (the other partial function of RLC) are in the central unit.

**Option 4 (RLC-MAC split)**

- MAC, physical layer and RF are in distributed unit. PDCP and RLC are in the central unit.

**Option 5 (intra MAC split)**

- RF, physical layer and some part the MAC layer (e.g. HARQ) are in the distributed unit. Upper layer is in the central unit.

**Option 6 (MAC-PHY split)**

- Physical layer and RF are in the distributed unit. Upper layers are in the central unit.

**Option 7 (intra PHY split)**

- Part of physical layer function and RF are in the distributed unit. Upper layers are in the central unit.

**Option 8 (PHY-RF split)**

- RF functionality is in the distributed unit and upper layer are in the central unit.

Editor's note: The options represented consist of a non-exhaustive list. The work in other working groups on protocols and functions definition shall be monitored and further split options based on such progress shall be added or removed if needed.

**Flexible functional split**

Some of the benefits of a New RAN architecture with the flexibility to split and move functions between central and distributed units are below:

- Flexible HW implementations allows scalable cost effective solutions
- A split architecture (between central and distributed units) allows for coordination for performance features, load management, real-time performance optimization, and enables NFV/SDN
- Configurable functional splits enables adaptation to various use cases, such as variable latency on transport

The NR design should support the flexibility to move RAN functions between the central unit and distributed unit, and should be studied.

### 6.1.2.2 Detailed Description of Candidate Split Options and Justification

#### 6.1.2.2.1 Option 1 (RRC/PDCP, 1A-like split)

#### 6.1.2.2.2 Option 2 (PDCP/RLC, 3C-like split)

Description: In this split option, RRC, PDCP are in the central unit. RLC, MAC, physical layer and RF are in the distributed unit.

Benefits and Justification: This option will allow traffic aggregation from NR and eLTE transmission points to be centralized. Additionally, it can facilitate the management of traffic load between NR and eLTE transmission points. Fundamentals for achieving a PDCP-RLC split have already been standardized for LTE Dual Connectivity, alternative 3C. Therefore this split option should be the most straightforward option to standardize and the incremental effort required to standardize it should be relatively small.

#### 6.1.2.2.3 Option 3 (High RLC/Low RLC Split)

Two approaches based on Real-time/Non Real-time function split are as follows:

**Option 3-1** Split based on ARQ

- Low RLC may be composed of segmentation and concatenation functions;
- High RLC may be composed of ARQ and re-ordering functions;

Description: This option splits the RLC sublayer into High RLC and Low RLC sublayers such that for RLC Acknowledge Mode operation, the ARQ and packet ordering functions may be performed at the High RLC sublayer residing in the central unit, while the segmentation may be performed at the Low RLC sublayer residing in the distributed unit.

Benefits and Justification:

- Compared to the PDCP-RLC (Option 2) split, this option has the advantage of being more robust under non-ideal transport conditions because the ARQ and packet ordering is performed at the central unit.
- This split option may also have better flow control across the split.
- Centralization gains: ARQ located in the CU provides more centralization or pooling gains over Option 2.
- The failure over transport network is also recovered using the end-end ARQ mechanism at CU. This provides protection for critical data and C-plane signaling.

- DUs without functions of RLC can handle more connected mode UEs as there is no RLC state information stored and hence no need for UE context.
- Reduced processing and buffer requirements in DU due to absence of ARQ protocol
- Could be used over multiple radio legs of different DUs for higher reliability (U-Plane and C-Plane)

Cons

- Comparatively, the split is more latency sensitive than the split with ARQ in DU, since re-transmissions are susceptible to transport network latency over a split transport network.

Overall, Option 3 where ARQ is located in CU provides significantly better pooling gains (packet processing) than Option 2. In addition, Option 2 requires larger packet buffers in DU. Therefore, it is beneficial to place ARQ function in CU according to the RAN function mapping shown in Option 3.

**Option 3-2** Split based on TX RLC and RX RLC

- Low RLC may be composed of transmitting TM RLC entity, transmitting UM RLC entity, a transmitting side of AM and the routing function of a receiving side of AM.
- High RLC may be composed of receiving TM RLC entity, receiving UM RLC entity and a receiving side of AM except the routing function.

##### 6.1.2.2.4 Option 4 (RLC-MAC split)

##### 6.1.2.2.5 Option 5 (intra MAC split)

##### 6.1.2.2.6 Option 6 (MAC-PHY split)

##### 6.1.2.2.7 Option 7 (intra PHY split)

Multiple realizations of this option are possible, including asymmetrical options in which one sub-option (e.g. 7-1) is used in the UL and another sub-option (e.g. 7-2) is used in the DL. A compression technique may be able to reduce the required transport bandwidth between the DU and CU.

In the UL, FFT, and CP removal reside in the DU. Two sub-variants are described below. Remaining functions reside in the CU.

In the downlink, iFFT and CP addition reside in the DU. Two sub-variants are described below. The rest of the PHY resides in the CU.

**Option 7-1**

In the UL, FFT, CP removal and possibly PRACH filtering functions reside in the DU, the rest of PHY functions reside in the CU. The details of the meaning of PRACH filtering are FFS.

In the DL, iFFT and CP addition functions reside in the DU, the rest of PHY functions reside in the CU.

**Option 7-2**

In the UL, FFT, CP removal, resource de-mapping and possibly pre-filtering functions reside in the DU, the rest of PHY functions reside in the CU. The details of the meaning of pre-filtering are FFS.

In the DL, iFFT, CP addition, resource mapping and precoding functions reside in the DU, the rest of PHY functions reside in the CU.

It is a requirement that both options allow the optimal use of advanced receivers. Whether or not these variants meets this requirement is FFS.

**Benefits and Justification for Option 7-1 and 7-2:**

- Compared to Option 8 this option is expected to reduce the fronthaul requirements in terms of throughput [details are FFS].

- It is expected to be able to maintain the ability to perform joint processing (both transmit and receive) across multiple TPs
- Compared to higher splits (i.e. options 1-4) the option has the advantage of supporting centralized scheduling, e.g. CoMP
- Option 7-1 allows the implementation of advanced receivers

##### 6.1.2.2.8 Option 8 (PHY-RF split)

### 6.1.2.3 Architectural and specification aspects

Editor's note: This chapter should at least handle the following questions: (1) How many splits will be specified and supported by open interfaces? (2) Will the tight LTE/NR interworking case effect the number of functional split options? (3) What is the granularity of the Centralized Unit – Distributed Unit functional split? (4) What is the reconfiguration dynamicity of the network functional split?.

### 6.1.2.4 Transport network aspects

#### 6.1.2.4.1 General

This section summarizes transport network requirements of different functional splits.

NOTE 1: It is understood that RAN3 has no intention to specify any transport network.

#### 6.1.2.4.2 Transport network requirements for an example New RAN architecture

According to TR 38.913 [5], the NR shall support up to 1GHz system bandwidth, and up to 256 antennas. A calculation relative to one of several possible transport deployments applied to a possible RAN architecture example shows that transmission between base band part and radio frequency part requires a bitrate over the transport network of about 614.4Mbps per 10MHz mobile system bandwidth per antenna port.

When the system bandwidth is increasing as well as the number of antenna ports, the required bitrate is linearly increasing. An example with rounded numbers is shown in the following table.

**Table 6.1.2.4.2-1 Examples of required bitrate on a transmission link for one possible PHY/RF based RAN architecture split**

| Number of Antenna Ports | Frequency System Bandwidth | | | |
|---|---|---|---|---|
| | 10 MHz | 20 MHz | 200 MHz | 1GHz |
| 2 | 1Gbps | 2Gbps | 20Gbps | 100Gbps |
| 8 | 4Gbps | 8Gbps | 80Gbps | 400Gbps |
| 64 | 32Gbps | 64Gbps | 640Gbps | 3200Gbps |
| 256 | 128Gbps | 256Gbps | 2560Gbps | 12800Gbps |

NOTE 1: Peak bitrate requirement on a transmission link = Number of BS antenna elements * Sampling frequency (proportional to System bandwidth) * bit width (per sample) + overhead.

### 6.1.3 UP-CP Separation

Editor's note: This section should capture an analysis on UP-CP separation based on identification of UP and CP functions, justification of the benefits in separating such functions into separate RAN entities and whether a dedicated RAN architecture for support of UP-CP separation is beneficial and feasible

#### 6.1.3.1 UP and CP Functions Description and Grouping

Editor's note: This section should capture a description of possible CP and UP functions, potential grouping of such functions taking into account possible RAN internal architectures and the benefits of such grouping schemes

#### 6.1.3.2 RAN architecture and interfaces for UP-CP Separation

Editor's note: This section should capture an analysis of whether it is beneficial and feasible to define specific RAN architectures and interfaces to support CP-UP separation

## 6.2 New RAN architecture

Editor's note: Intention is to capture overall New RAN architecture.

## 6.3 Interfaces

Editor's note: Intention is to capture protocol stacks and list of functions for each agreed interfaces in the New RAN architecture.

### 6.3.1 General

The following options of [7] for providing NR access to suitably capable UEs should be considered in discussions on the RAN-CN interface, and the interface between (Evolved) E-UTRA and NR RAT.

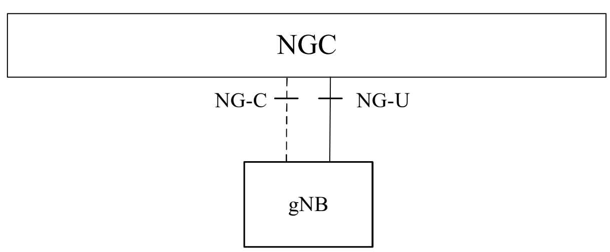



**Figure 6.3.1-1: Option 2**

In Option 2, the gNB is connected to the NGC.

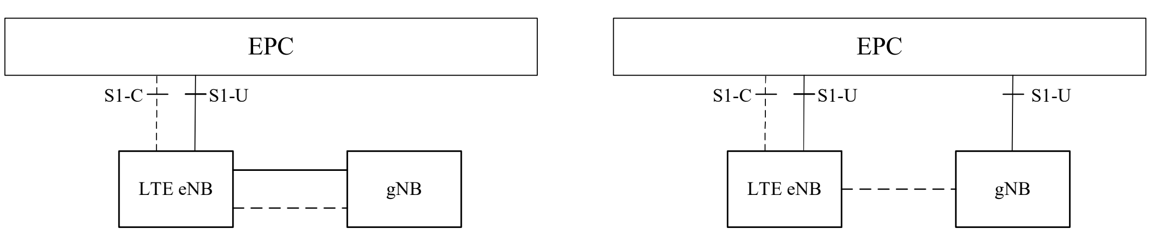


**Figure 6.3.1-2: Options 3 and 3A**

In Option 3/3A, the LTE eNB is connected to the EPC with Non-standalone NR.  The NR user plane connection to the EPC goes via the LTE eNB (Option 3) or directly (Option 3A).

Editor's note: Terminology related to Option 3/3A can be further discussed, if needed.

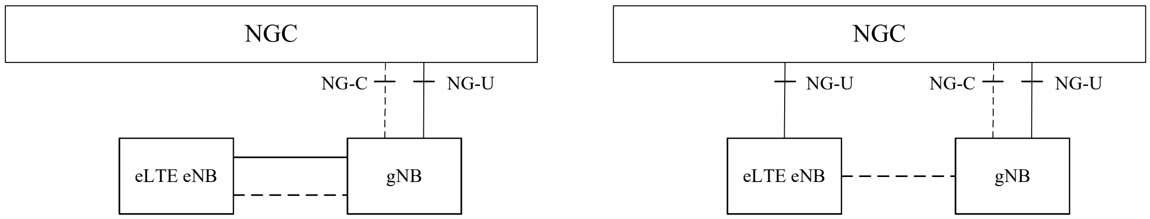



**Figure 6.3.1-3: Options 4 and 4A**

In Option 4/4A, the gNB is connected to the NGC with Non-standalone Evolved E-UTRA.  The Evolved E-UTRA user plane connection to the NGC goes via the gNB (Option 4) or directly (Option 4A).

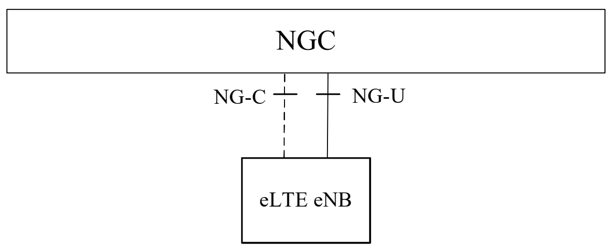



**Figure 6.3.1-4: Option 5**

In Option 5, the eLTE eNB is connected to the NGC.

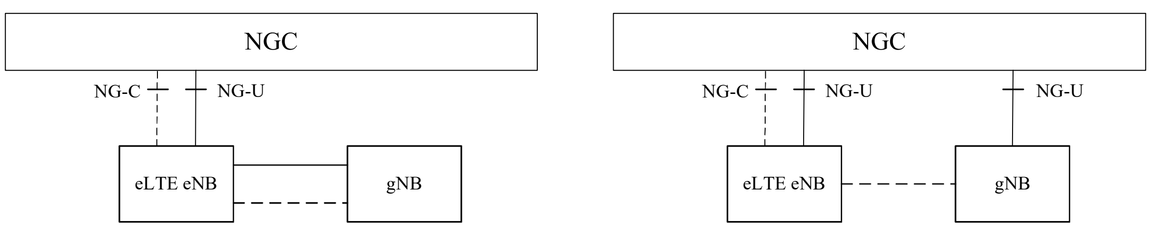



**Figure 6.3.1-5: Options 7 and 7A**

In Option 7/7A, the eLTE eNB is connected to the NGC with Non-standalone NR.  The NR user plane connection to the NGC goes via the eLTE eNB (Option 7) or directly (Option 7A).

## 6.3.2 RAN-CN interface

### 6.3.2.1 RAN-CN interface connectivity scenarios

In order to support the options described in section 6.3.1, the following scenarios for connectivity between  RAN consisting of E-UTRA, Evolved E-UTRA and NR, and a CN consisting of an NGC and an  EPC should be considered in the discussions on RAN-CN interface definition . The connectivity scenario in figure 6.3.2.1-1 includes support for Option 3/3A, while figure 6.3.2.1-2 includes support for Option 2, Option 4/4A, Option 5, and Option 7/7A.

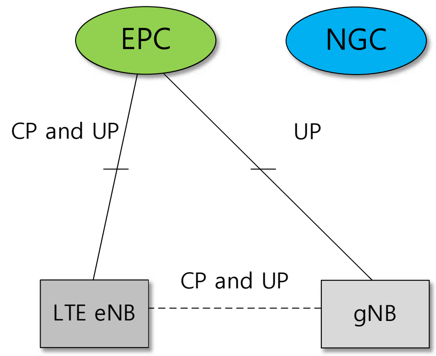



**Figure 6.3.2.1-1: E-UTRA and NR connected to the EPC. The CP between EPC and gNBis FFS, but any further discussion in RAN3 is dependent on RAN Plenary decision.**

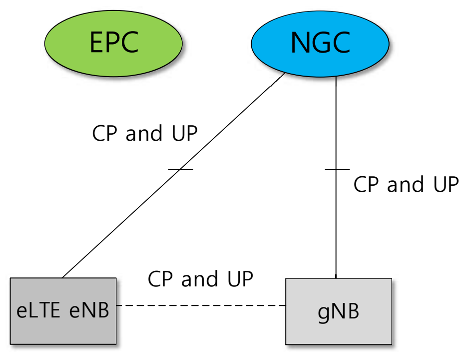



**Figure 6.3.2.1-2: Evolved E-UTRA and NR connected to the NGC (Note: In this scenario, eLTE eNB and gNB can be collocated.).**

### 6.3.2.2 General principles

The general principles for the specification of the NG1 interface are as follows:

- the NG1 interface shall be open;
- the NG1 interface shall support the exchange of signalling information between the NR BS and 5G CN;
- from a logical standpoint, the NG1 is a point-to-point interface between an NR BS within the RAN and an 5G CN node. A point-to-point logical interface shall be feasible even in the absence of a physical direct connection between the NR BS and 5G CN;
- the NG1 interface shall support contral plane and user plane separation;
- the NG1 interface shall separate Radio Network Layer and Transport Network Layer;
- the NG1 interface shall be future proof to fulfil different new requirements and support of new services and new functions;
- the NG1 interface shall be decoupled with the possible RAN deployment variants.
- the NG1 Application Protocol shall support modular procedures design and use a syntax allowing optimized encoding /decoding efficiency.

NOTE 1: The working assumption is that the interface between eLTE eNB and 5G CN is also NG1.

#### 6.3.2.3 NG Interface Functions

NG-C interface supports following functions:

- Interface management: procedures to manage the NG-C interface;
- UE context management: procedures to manage the UE context between the RAN and CN;
- UE mobility management: procedures to manage the UE mobility for connected mode between the RAN and CN;
- Transport of NAS messages: procedures to transfer NAS messages between the CN and UE;
- Paging: procedures to enable the CN to generate Paging messages sent to the RAN and to allow the RAN to page the UE in RRC_IDLE state;
- PDU Session Management (FFS): procedures to establish, manage and remove PDU sessions that are made of data flows carrying UP traffic.

### 6.3.3 RAN internal interface

#### 6.3.3.1 Xn Interface

The interface allowing to interconnect two gNBs or one gNB and one eLTE eNB with each other is referred to as the Xn interface.

Editor's note: The applicability of this interface for a connection between two eLTE eNBs is FFS.

##### 6.3.3.1.1 General principles

The general principles for the specification of the Xn interface are as follows:

- the Xn interface shall be open;
- the Xn interface shall support the exchange of signalling information between the endpoints, in addition the interface shall support data forwarding to the respective endpoints;
- from a logical standpoint, the Xn is a point-to-point interface between the endpoints. A point-to-point logical interface should be feasible even in the absence of a physical direct connection between the endpoints.
- the Xn interface shall support control plane and user plane separation;
- the Xn interface shall separate Radio Network Layer and Transport Network Layer;
- the Xn interface shall be future proof to fulfil different new requirements, support new services and new functions.

# 7 Radio access network procedures for New RAT

Editor's note: Intention is to capture key procedures (e.g. mobility) including message sequences. Interface procedures between RAN and UE can be touched upon to provide a holistic view, but focus in this TR should be more on RAN procedures. Sub-sections can be introduced (per procedure) as discussion progresses.

## 7.1 Tight interworking between new RAT and LTE

### 7.1.1 General

Option 3/3a, 4/4a and 7/7a of deployment scenarios can be considered as LTE-NR tight interworking.

In Option 3/3a, Dual Connectivity (DC) specified in TS 36.300 [12] and relevant stage 3 specifications (e.g., TS 36.423 [13]) should be reused as baseline considering the fact that EPC should not be impacted. Therefore, for the Xx interface between LTE eNB and gNB, the procedures and protocols would remain alike those of DC, while minor enhancements might not be ruled out.

Editor's note: Description for Option 4/4a and 7/7a is FFS.

### 7.1.2 Option 3/3a

#### 7.1.2.1 Architectural aspects

Assuming DC specified in TS 36.300 [12] as the baseline for LTE-NR tight interworking in this option, Radio Protocol Architecture for the User Plane can be defined in Figure 7.1.2.1-1. LTE eNB and gNB are assumed to have the role similar to MeNB (Master eNB) and SeNB (Secondary eNB) specified in TS 36.300 [12], respectively. NR PDCP is one of the NR sub-layers to handle SDUs of the S1-U interface as well as the SDUs of the NGC interface into different DRBs according to the QoS information associated with the SDU.

Editor's note: The below figure is an example. NR PDCP, NR RLC and NR MAC are pending to RAN2.

Editor's note: Support of new bearer type is FFS.

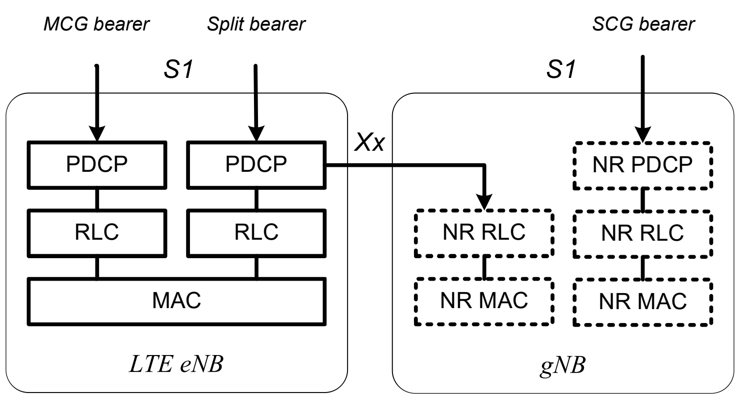



**Figure 7.1.2.1-1: Radio Protocol Architecture for split bearer and SCG bearer in Option 3/3a**

Network interface configurations can be defined in Figure 7.1.2.1-2 and Figure 7.1.2.1-3.

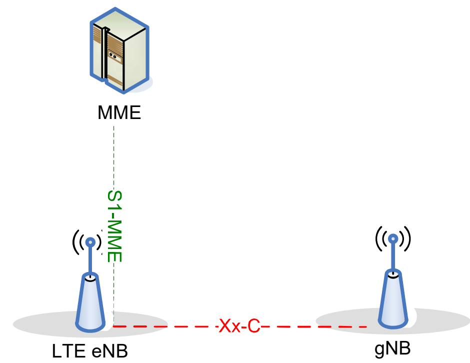



**Figure 7.1.2.1-2: C-Plane connectivity for Option 3/3a**

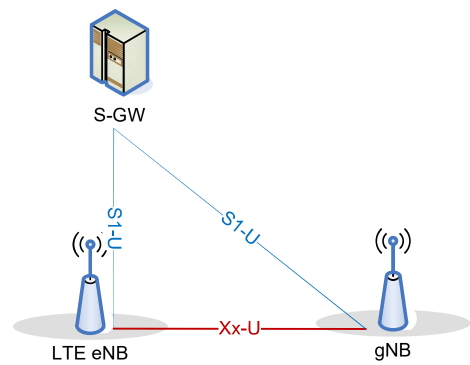



**Figure 7.1.2.1-3: U-Plane connectivity for Option 3/3a**

### 7.1.2.2 Procedural aspects

The procedures defined under section 10.1.2.8 (Dual Connectivity operation) in TS 36.300 [12] are listed below. In this list, LTE eNB and gNB connected via Xx are considered to have the role similar to MeNB (or eNB without DC) and SeNB, respectively. The DC procedures defined in TS 36.423 [13] should be used as baseline.

- SeNB Addition
- SeNB Modification (MeNB initiated SeNB Modification)
- SeNB Modification (SeNB initiated SeNB Modification)
- Intra-MeNB handover involving SCG change
- SeNB Release (MeNB initiated SeNB Release)
- SeNB Release (SeNB initiated SeNB Release)
- Change of SeNB
- MeNB to eNB Change
- SCG change
- eNB to MeNB change
- Inter-MeNB handover without SeNB change

- Addition of a hybrid HeNB as the SeNB

Editor's note: Applicability of the above-listed DC procedures is FFS.

## 7.2 Standalone new RAT operation

### 7.2.1 Mobility

#### 7.2.1.1 Intra-RAT mobility scenarios

**Scenario: Inter NR BS mobility with interface**

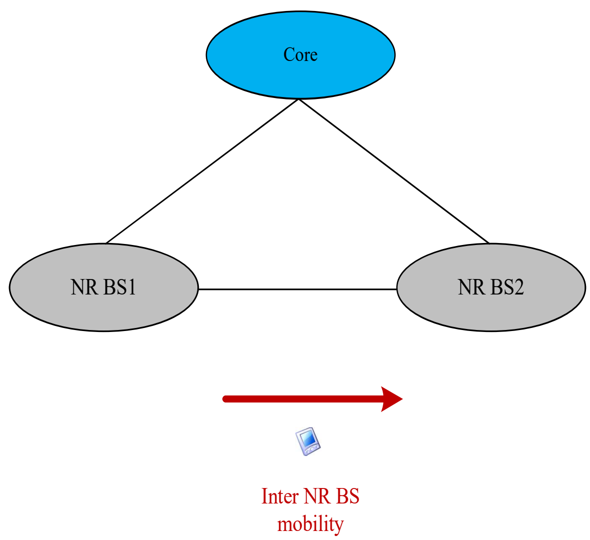



**Figure 7.2.1.1-1: Inter NR BS mobility scenario**

Editor's note: Intra NR BS mobility scenario is subject to discussion with more progress from RAN2 and RAN3 on functional split and RAN architecture.

#### 7.2.1.2 Inter-RAT handover with LTE

There are three inter-RAT handover with LTE scenarios as shown in Figure 7.2.1.2-1.

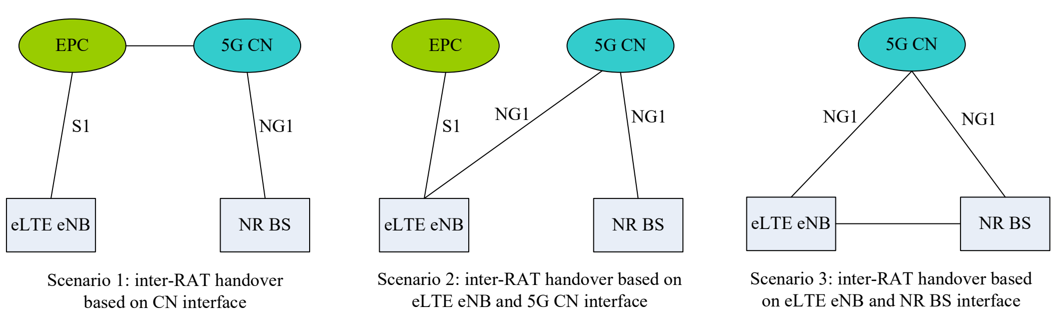



**Figure 7.2.1.2-1: Inter-RAT handover with LTE scenarios**

NOTE 1: The interface between EPC and 5G CN is pending to SA2.

NOTE 2: Further justification on Scenario 2 is needed.

# 8 Realization of RAN Network Functions

Editor's note: Intension is to capture aspects related to virtualization in this section.

## 8.1 Requirements

NR shall allow Centralized Unit (CU) deployment with Network Function Virtualization (NFV).

Editor's note: the definition of RAN Network Functions, Virtualization and CU are FFS.

# 9 Realization of Network Slicing

## 9.1 Key principles for support of Network Slicing in RAN

Network Slicing is a new concept to allow differentiated treatment depending on each customer requirements. With slicing, it is now possible for Mobile Network Operators (MNO) to consider customers as belonging to different tenant types with each having different service requirements that govern in terms of what slice types each tenant is eligible to use based on Service Level Agreement (SLA) and subscriptions.

The following key principles apply for support of Network Slicing in RAN

**RAN awareness of slices**

- RAN shall support a differentiated handling of traffic for different network slices which have been pre-configured. How RAN supports the slice enabling in terms of RAN functions (i.e. the set of network functions that comprise each slice) is implementation dependent.

Editor's note: It is still FFS if 3GPP will still additionally want to standardize a few basic slices and the network functions that comprise them (e.g. eMBB, Massive MTC).

**Selection of RAN part of the network slice**

- RAN shall support the selection of the RAN part of the network slice, by a slice ID provided by the UE which unambiguously identifies one of the pre-configured network slices in the PLMN.

Editor's note: How the UE gets this unambiguous slice ID is FFS and to be decided with SA2. The ID could be sent to the UE by the CN after the CN has selected the slice (e.g. similar to eDECOR feature) or it could be pre-configured in the UE.

Editor's note: it is FFS how the RAN verifies that the UE is authorized to select the slice and when this verification happens.

Editor's note: It is FFS if the RAN may also select the slice based on specific resources accessed by the UE.

**Resource management between slices**

- RAN shall support policy enforcement between slices as per service level agreements. It should be possible for a single RAN node to support multiple slices. The RAN should be free to apply the best RRM policy for the SLA in place to each supported slice.

Editor's note: How RAN handles the requirements coming from the service level agreements is to be discussed with SA2.

**Support of QoS**

- RAN shall support QoS differentiation within a slice.

Editor's note: It is FFS if RAN shall additionally support QoS enforcement independently per slice.

**RAN selection of CN entity**

- RAN shall support initial selection of the CN entity for initial routing of uplink messages based on received slice ID and a mapping in the RAN node (CN entity, slices supported). If no slice ID is received, the RAN selects the CN entity based on NNSF like function, e.g. UE temporary ID.

**Resource isolation between slices**

- RAN shall support resource isolation between slices. RAN resource isolation may be achieved by means of RRM policies and protection mechanisms that should avoid that shortage of shared resources in one slice breaks the service level agreement for another slice. It should be possible to fully dedicate RAN resources to a certain slice

Editor's note: Resource isolation needs to be clarified: It is unclear if resource isolation would imply that multiple slices cannot share control plane (respectively user plane) resources or processing resources in common. It is unclear if resource isolation would imply that cryptographic means should be used to isolate CP and UP traffic between slices.

## 9.2 Solutions for selection of Network slice and CN entity by gNB

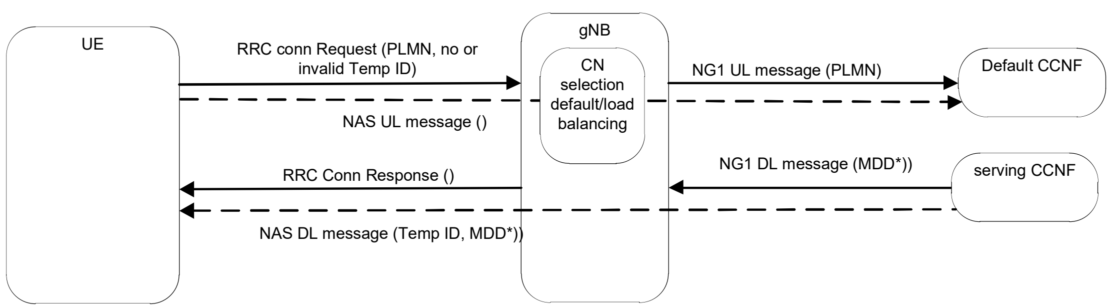
### 9.2.1 Solution 1

Editor's note: this section presents a RAN solution for network slice selection and CN entity selection which is based on the solution described in section 6.1.2 of TR 23.799 [6]. This solution enables support of multiple slices simultaneously supported by a UE.

In this solution the Network Slice and the CN entity are selected based on a Multi-Dimensional Descriptor called MDD as described in TR 23.799 [6]. In this case the CN entity is a common control plane entity in NG Core (CCNF) – common to all supported slices - as described in section 6.1.2 of TR 23.799 [6].

The RAN selection of the Network slice and the selection of the appropriate CCNF can be partitioned into the following five cases.

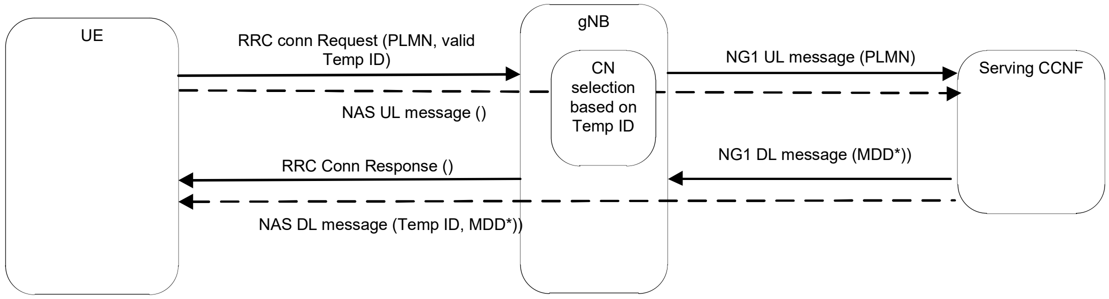
Selection case 1: no MDD, no or invalid Temp ID




Valid temp ID means valid for this PLMN.

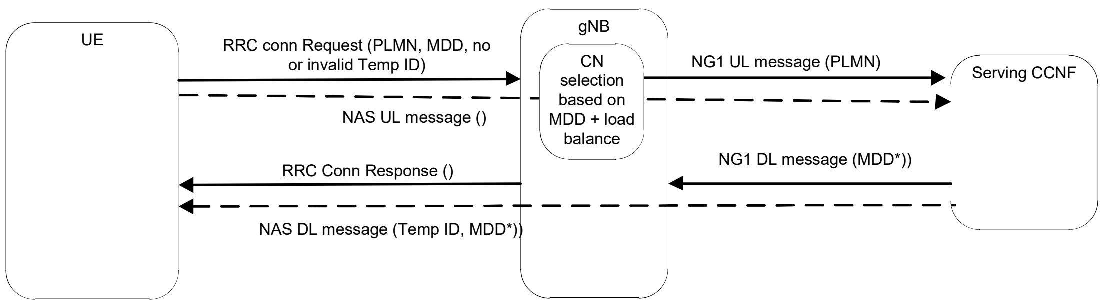
In absence of Temp ID information valid for the indicated PLMN in the RRC connection request, the gNB needs to route to a default configured CCNF. After validation steps, the Default CCNF, depending on UE capabilities and subscription information, will select a suitable serving CCNF. This serving CCNF could be the default CCNF itself. A Temp ID corresponding to serving CCNF is sent back to the UE at NAS level. In case an MDD applies for the selected PLMN, this MDD is sent back to the UE together with the Temp ID at NAS level. The MDD is then also indicated in the NG1 DL message to gNB.

Selection case 2: no MDD, valid Temp ID



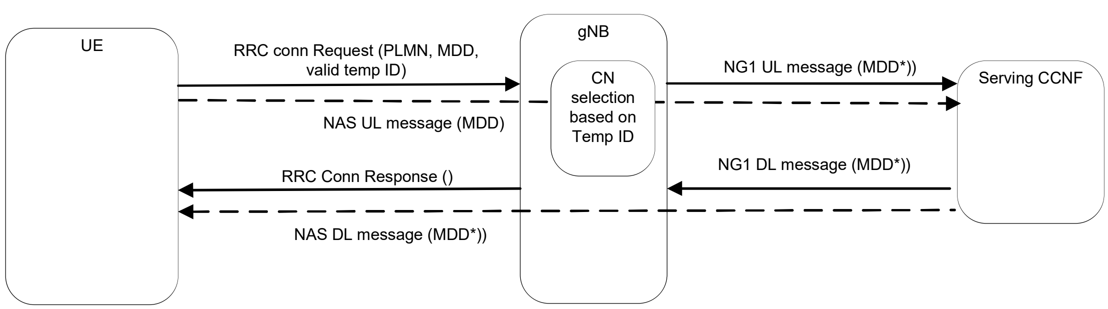

If a Temp ID information valid for the indicated PLMN is included in RRC Connection Request, the gNB routes based on Temp ID to the serving CCNF. In case an MDD applies for the selected PLMN, this MDD is sent back to the UE at NAS level. The MDD is then also indicated in the NG1 DL message to gNB.

Selection case 3: MDD, no or invalid Temp ID




In absence of Temp ID information valid for the indicated PLMN in the RRC connection request, the gNB selects a serving CCNF based on the MDD when present. If multiple CCNF are available and suitable for the MDD, gNB can apply load balancing criteria. RAN also uses the included MDD for RAN slice awareness (slice specific behavior, congestion control, resource isolation). After validation steps, a Temp ID corresponding to serving CCNF is sent back to the UE at NAS level. The MDD is sent back to the UE together with the Temp ID at NAS level to either reconfirm the initially requested MDD or indicates a new one if policy has changed. The MDD is also indicated in the NG1 DL message to gNB for confirmation.

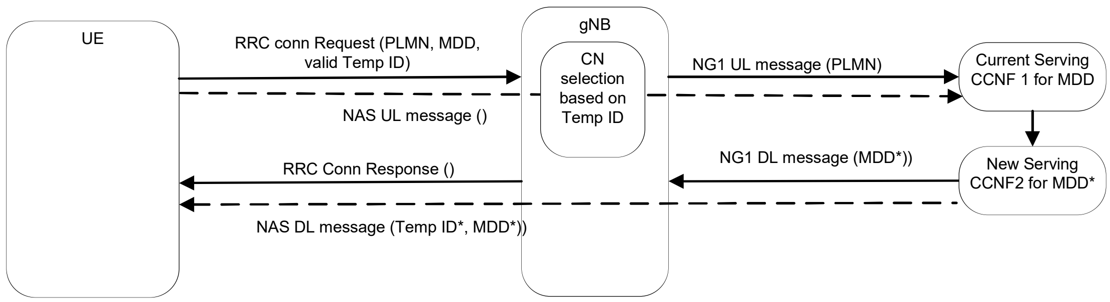
Selection case 4: MDD, valid Temp ID




›p2

If a Temp ID information valid for the indicated PLMN is included in RRC Connection Request, the gNB routes based on Temp ID to the serving CCNF regardless of MDD presence. RAN uses the included MDD for RAN slice awareness (slice specific behavior, congestion control, resource isolation). The MDD is sent back to the UE with the Temp ID at NAS level to either reconfirm the initially requested MDD or indicates a new one if policy has changed . The MDD is then also indicated in the NG1 DL message to gNB for confirmation.

Selection case 5: Modification of MDD, valid Temp ID




›p2

If a Temp ID information valid for the indicated PLMN is included in RRC Connection Request, the gNB routes based on Temp ID to the serving CCNF regardless of MDD presence. RAN uses the included MDD for RAN slice awareness (congestion control, resource isolation). If the current MDD needs to be updated into MDD* because some of the MDD vectors are no more supported (i.e. operator policy has changed for the UE due to UE capabilities e.g. the subscriber may use different types of UEs by swapping the UICC) and if CCNF1 is not compatible with the new MDD*, the NG Core may select a new CCNF2 compatible with MDD*. In this case the MDD* is sent back from CCNF2 to the UE together with a new Temp ID* corresponding to CCNF2 at NAS level. The new MDD* is then also indicated to gNB over NG1 DL message.

## 9.2.2 Solution 2

Network Slice selection may be realised in at least two mechanisms. One possible way is to configure the UE with a list of slice IDs to which it is allowed to access. Another way is to let the CN notify the UE of the Slice IDs it can access and let the UE rely on such information onwards.

The first case of UE configured Slice ID is made of the following general steps:

- The UE will be configured with Slice IDs and with criteria to present the right Slice ID to the RAN.
- When the UE requests to access the RAN it will present a Slice ID as per criteria previously configured.
- By receiving the Slice ID, the RAN shall be able to
    - Identify the policies that apply to the selected slice and assign RAN resources accordingly.
    - Identify the CN node that supports the slice ID presented by the UE.

In order to achieve the CN node selection the RAN would need to be informed about the slice IDs supported by connected CN nodes. This can be achieved by letting RAN and CN exchange information on supported slice IDs at RAN-CN interface setup.

Figure 9.2.1-1 summarises the steps described in the case of Slice ID provided by the UE.

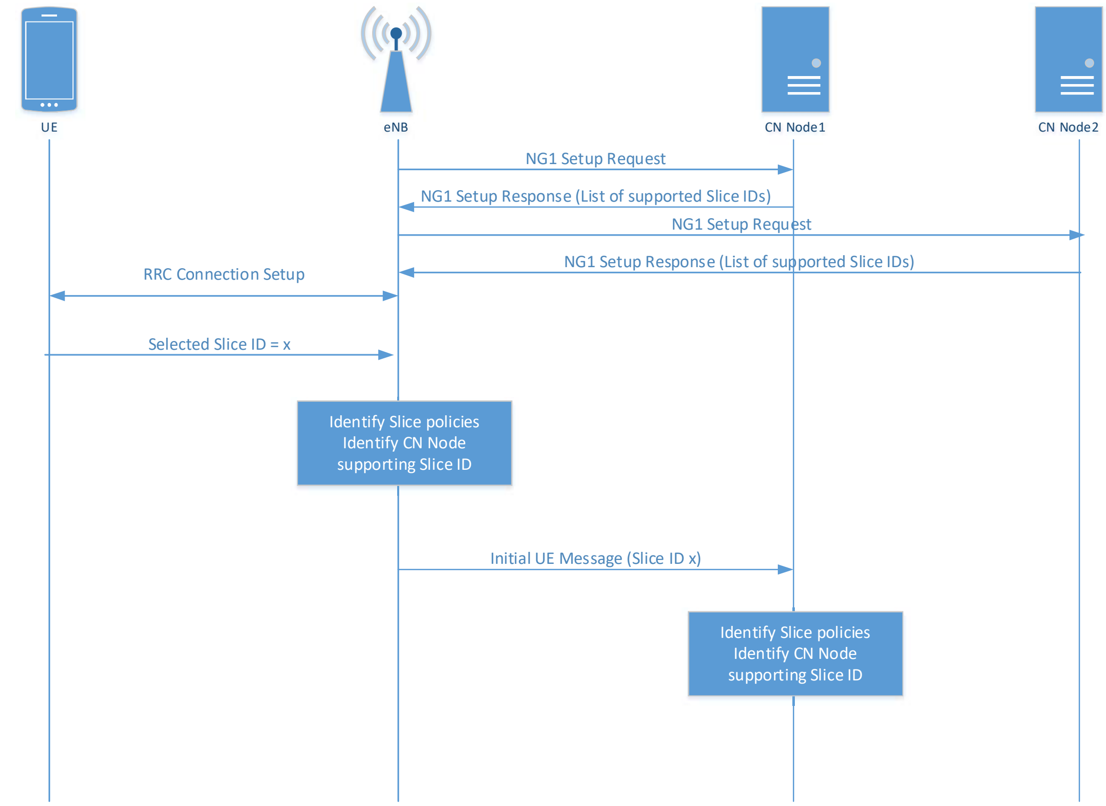



**Figure 9.2.1-1: Slice selection when Slice ID is provided by the UE to RAN**

The second case, where the slice ID is not initially provided by the UE and it is configured by the CN to the UE, consists of the following general steps:

- The UE presents a NAS Service Request with no specific Slice ID.
- The serving RAN routes the NAS PDU to a default CN Node.
- The default CN node analyses the NAS request and retrieves UE subscriber information. Based on the UE subscriber information and NAS Service Request the default CN Node reroutes the request to a different CN Node. The latter could be achieved by reusing procedures similar to existing S1: Reroute NAS Request.
- The new CN Node (reached via reroute) responds to the RAN with an assigned Slice ID for the UE and it responds to the UE with a NAS PDU in which usable Slice IDs are provided.
- From this moment onwards the UE will present to the RAN a slice ID that intends to access at RRC connection establishment.

Figure 9.2.1-2 summarises the steps described in the case of Slice ID configured by the CN to the UE.

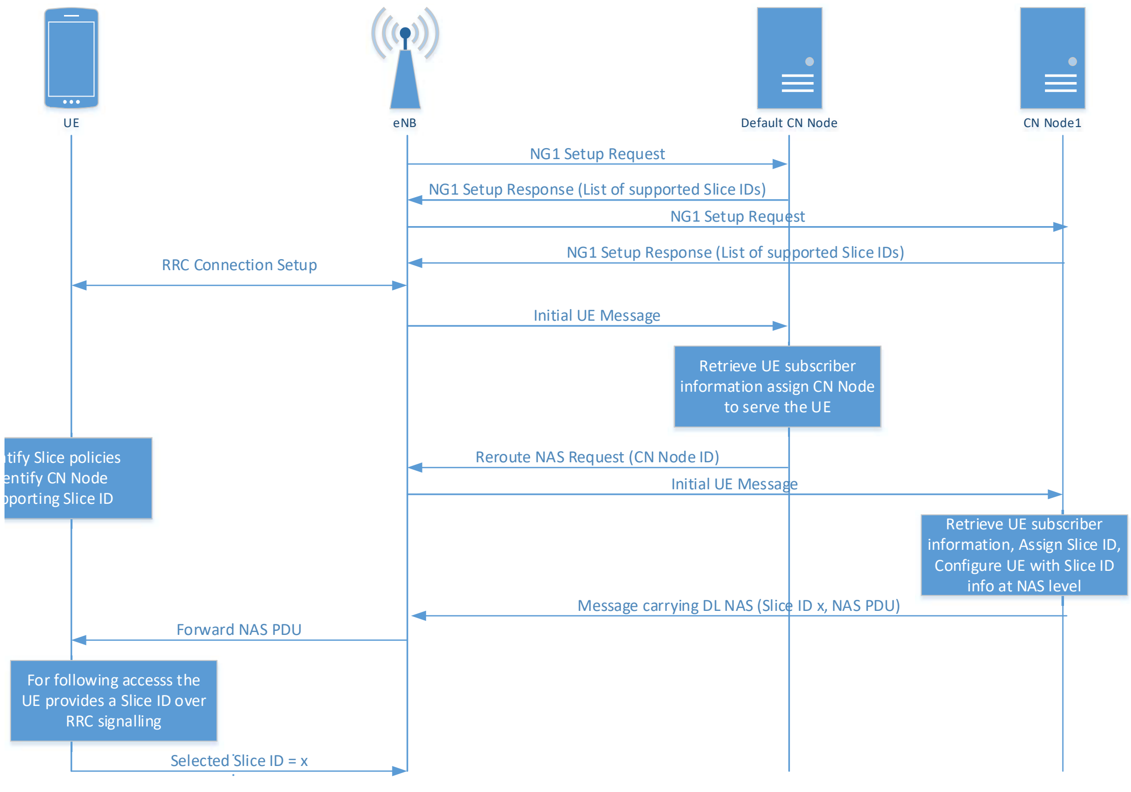



**Figure 9.2.1-2: Slice selection when Slice ID is provided by the CN**

In both mechanisms described above it is assumed that a UE can only access in parallel the network slices supported by the CN node serving it. Namely, a UE cannot access multiple slices served by different CN nodes at the same time.

### 9.2.3 Solution XXX

Editor's note: this section presents a RAN solution xxx.

## 9.3 Resource Isolation between slices

Resource isolation enables specialized customization and avoids one slice affecting another slice. Hardware/software resource isolation is up to implementation. Each slice may be assigned with either shared or dedicated radio resource up to RRM implementation and SLA.

When assigned dedicated radio resource the slice may be isolated and configured by RAN with one or more of below items:

- Time/frequency/code resources etc;
- Access channel;

Editor's note: It is up to RAN1/RAN2 to decide how to partition access channel e.g. in frequency, time and preamble.

- Independent Access control, Load control, QOS etc.

Logically, slices may be isolated in terms of DRBs.

The RAN should be allowed to serve traffic for different slices via shared resources.

# 10 Ultra-Reliable and Low Latency Communications

NR supporting Ultra-Reliable and Low Latency Communications may support end-to-end latency of 1ms as specified in in TR 22.862 [4]. End-to-end latency is defined as the latency between the UE and the application server.

Editor's note: RAN3 will work on NR RAN architecture solutions to support end-to-end latency requirements in coordination with SA2.

Editor's note: In this section RAN3 will capture NR RAN architecture requirements and solutions to support end-to-end latency requirements defined in TR 22.862 [4].

# 11 QoS

Editor's notes: Capture QoS related aspects.

# 12 SON

Editor's notes: Capture SON related aspects.

## 12.1 Scope of SON for New RAT

Self-organizing network (SON) functions such as support for mobility robustness optimization, mobility load balancing, RACH optimization, energy savings, etc. were introduced in LTE to make deployment easier and cheaper. Equivalent functionality shall also be considered for the New RAT. In addition, new SON functionality may be considered, taking into account new NR features, new use cases supported by NR, and the operation in complex multi-RAT, multi-band and multi-vendor deployments.

## 12.2 Self configuration procedures

### 12.2.1 Automatic neighbour relations

Relations in 5G can be between entities, supporting different radio resource management procedures. Such procedures may include coordination of resources and identifiers, mobility, load and traffic sharing, multi-connectivity, etc. An example of such neighbour relation is relation among gNBs and eLTE eNBs interconnected with Xn.

# 13 Wireless relay

Editor's notes: Capture wireless relay related aspects.

## 13.1 Scenarios

The following scenarios for wireless relay in 5G RAN should be considered.

- Single-hop stationary relay
  - A relay node may connect to a donor node through wireless backhaul to extend network coverage. Relay nodes could be used for network densification and coverage extension with a trade-off between deployment cost, deployment flexibility, capacity loss, and increased latency at the donor with respect to other solutions.

- Multiple-hop relay
  - Due to the limited coverage of a relay node, it may be needed to consider the support for multiple hop relay to extend network coverage. In this case, the traffic may be transmitted via one or more intermediate relay nodes, i.e. hop by hop. Multi hop relay may apply to some practical use cases (e.g. street canyon), with a trade-off between deployment cost, deployment flexibility, capacity loss, and increased latency at the donor with respect to other solutions.
- Multiple donor relay
  - To further improve the bandwidth, a relay node may connect to multiple donors. The relevance of this scenario with respect to e.g. small cell and dual connectivity use cases, should be analysed.
- Mobile relay:
  - A relay node may be deployed on a vehicle, and provides wireless connectivity service to end user inside the vehicle. The relay node's donor node may be changed, e.g. moving across the coverage of different donor node. The relevance of this scenario with respect to previous studies in LTE should be considered; furthermore, the feasibility of this scenario with respect to the physical layer should be evaluated by the appropriate WG(s).

# 14 Migration towards RAN for New RAT

Editor's notes: Intention is to capture how migration from existing E-UTRAN to the RAN supporting new RAT may occur (i.e. to capture the required modification/upgrades to E-UTRAN in order to evolve to the RAN supporting new RAT).

## 14.1 Potential migration path 1 [8]

### 14.1.1 Step 1 deployment

When NR is launched on day 1, a likely scenario is to deploy NR on frequencies higher than those being used for LTE. In this case, the NR coverage is most likely to be much smaller than the existing LTE coverage, especially for frequencies above 6 GHz. For eMBB which can be regarded as continuous evolution of the existing cellular service, it is desirable if the existing LTE coverage can be leveraged to provide nationwide continuous coverage and mobility. In addition to that, the NR coverage enables to boost U-plane capacity in the target stop area where the traffic load is high. LTE-NR Dual Connectivity enables operators to launch the NR service as such; eNB acts as MeNB and gNB acts as SeNB. Since LTE eNB as MeNB is already connected to EPC, leveraging EPC can further drives cost effective and early launch of the NR service for eMBB. LTE-NR Dual Connectivity via EPC.

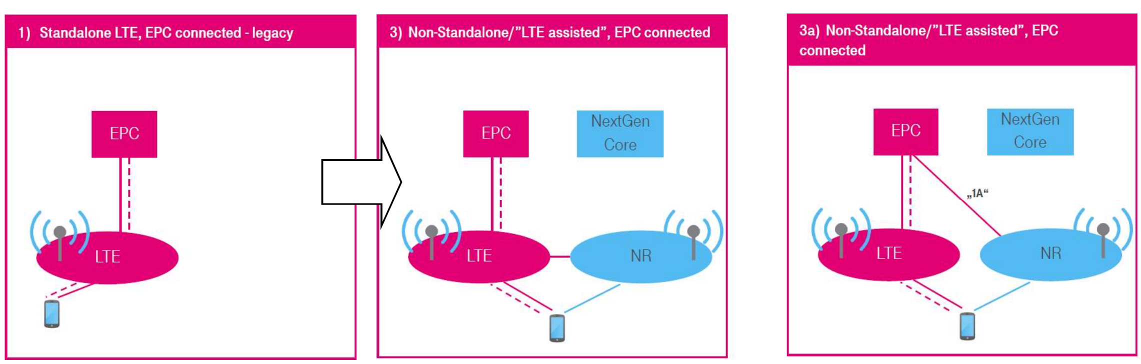



**Figure 14.1.1-1: Migration path from LTE standalone to LTE-NR DC via EPC**

### 14.1.2 Step 2 deployment and on wards

After the day 1 deployment, next step in question is how NextGen Core is introduced. As for migration from EPC, the following three roles can be considered.

- NextGen Core is evolution of EPC.
- NextGen Core encompasses EPC as a slice.
- NextGen Core replaces EPC.

From operator's viewpoints, it has to be understood whether:

a) NextGen Core is backward compatible with EPC and so can provide the same service as EPC can.

b) NextGen Core is not backward compatible with EPC and so is designed to provide a new service and use case which cannot be done by EPC.

Even after the day 1 NR deployment and the existing LTE coverage is replaced with NR, operators can reap the benefit if EPC can accommodate NR as a standalone RAT. Even though NextGen Core is introduced, NextGen Core might work as EPC as explained above. As such, NextGen Core in relation to EPC has to be clarified before making a final decision on the RAN-CN interface scenarios to be standardised.

## 14.2 Potential migration path 2 [9]

One identified sequence of migration from existing E-UTRAN to the New RAN is as follows:

Step 1: Early 5G Deployment utilizing Option 3

Step 2: Migration to Option 7 – Includes Simultaneous Support for Option 3

Step 3: Migration to Option 4 and/or 2 – Plus Simultaneous Support for Option 3 & Option 7

NOTE: Options 2, 3, 4 and 7 are described in section 6.3.1.

- Option 3 may include 3 and/or 3a.
- Option 4 may include 4 and/or 4a.
- Option 7 may include 7 and/or 7a.

### 14.2.1 Mobility and Service Continuity between NR and (Evolved) E-UTRA

The applicable mobility and service continuity scenarios should be evaluated for each migration step.

Editor's note: Intention for the text below is to establish a placeholder to begin to capture a description of service continuity mobility scenarios that are required to be supported during migration steps. It is FFS on whether every mobility and service continuity applies to each migration path/step and how to capture this.

The following mobility and service continuity scenarios may be supported during operation of Steps 1-3 above:

1. [e.g. Data Mobility between Evolved E-UTRA and NR with service continuity for data - FFS];
2. [e.g. Voice Mobility between Evolved E-UTRA and NR with service continuity for voice] - FFS]
3. [e.g. Data Mobility between E-UTRA and NR with service continuity for data – FFS]
4. [e.g. Voice Mobility between E-UTRA and NR with service continuity for voice - FFS]
5. [e.g. Other mobility scenarios - FFS]

## 14.3 Potential migration path 3 [10]

The migration paths from existing E-UTRAN to RAN for new RAT are based on the deployment architecture set options considered in [7]. The potential migration paths are as follow.

### 14.3.1 Potential migration path 3-1

LTE/EPC -> Option 2 + Option 5 -> Option 4/4a -> Option 2

- Step 1: Option 2 and 5 are deployed in parallel once the NG core is available. Evolved LTE assisted eLTE-NR tight interworking (Option 7&7a) can also be supported at this step.
- Step 2: NR assisted eLTE-NR tight interworking could be deployed
- Step 3: Only NR and NG core are used.

UE mode:

- Dual mode UE in step 1 with NG NAS.
- Only NR NAS in step 2
- NR UE in step 3

### 14.3.2 Potential migration path 3-2

LTE/EPC -> Option 2 + Option 5 -> Option 2

- Step 1: Option 2 and 5 are deployed in parallel once the NG core is available. Evolved LTE assisted eLTE-NR tight interworking (Option 7&7a) can also be supported at this step.
- Step 2: Only NR and NG core are used.

UE mode:

- Dual mode UE in step 1 with NG NAS
- NR UE in step 2

### 14.3.3 Potential migration path 3-3

LTE/EPC -> Option 3/3a -> Option 4/4a-> Option 2

- Step 1: Non-standalone NR is deployed for support of LTE-NR tight interworking, based on legacy LTE network. Option 3&3a can be applied.
- Step 2: Standalone NR is deployed together with the NG core. NR assisted eLTE-NR tight interworking could also be deployed.
- Step 3: Only NR and NG core are used.

UE mode:

- Only LTE NAS in step 1
- Only NR NAS in step 2
- NR UE in step 3

### 14.3.4 Potential migration path 3-4

LTE/EPC -> Option 7/7a -> Option 2

- Step 1: LTE is evolved to eLTE to connect the NG core and non-standalone NR is deployed for support of eLTE-NR tight interworking
- Step 2: Only NR and NG core are used.

UE mode:

- Only NR NAS in step 1
- NR UE in step 2

# 15 Interworking with non-3GPP systems

Editor's notes: Capture non-3GPP interworking related aspects.

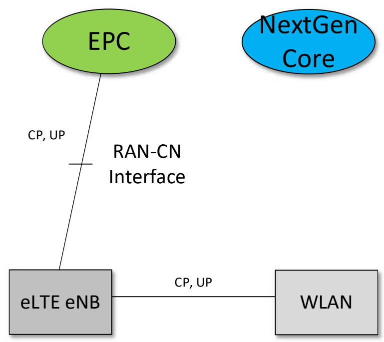



**Figure 15-1: LTE connected to the EPC, WLAN interworking with LTE via inter node interface. In this scenario it is assumed that there is an interface between LTE and WLAN.**

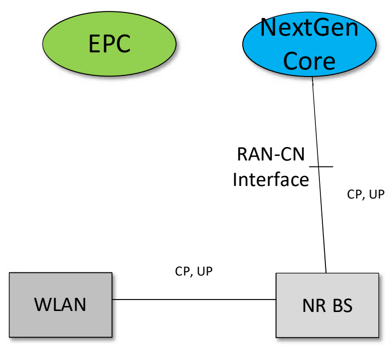



**Figure 15-2: NR connected to the NextGen Core, WLAN interworking with NR via inter node interface. In this scenario it is assumed that there is an interface between NR BS and WLAN**

# Annex A: Transport network and RAN internal functional split

When considering functional split options, the following transport performance requirements may be expected. The values given in the table are informative and for reference. The following transport characteristics deemed to be relevant:

1) Transport latency

2) Transport bandwidth

Those transport characteristics are contributing finally to deployment costs.

On the other hand, certain features and/or use cases like Ultra Reliable and Low Latency communication (URLLC) may require a certain split to support the features and/or use cases.

The following Table A-1 is proposed to be maintained during the SI while the knowledge about the protocol stack for NR and the related requirements on the transport are evolving.

**Table A-1 Requirements on the underlying transport network due to a certain functional split, as a consequence to support a certain feature/use case**

| Protocol Split option[1] | Required bandwidth | Max. allowed one way latency [ms] | Delay critical feature[2] | Comment |
|---|---|---|---|---|
| Option 1 | [*DL: 4Gb/s*]<br>[*UL: 3Gb/s*] | [*10ms*] | | |
| Option 2 | [*DL: 4016Mb/s*]<br>[*UL:3024 Mb/s*] | [*1.5~10ms*] | | |
| Option 3 | [*lower than option 2 for UL/DL*] | [*1.5~10ms*] | | |
| Option 4 | [*DL:5226.7Mb/s*]<br>[*UL:4500Mb/s*] | [*approximate 100us*] | | |
| Option 5 | [*DL: 5626.7Mb/s*]<br>[*UL: 7140 Mb/s*] | [*hundreds of microseconds*] | | |
| Option 6 | [*DL: 5626.7Mb/s*]<br>[*UL:7140 Mb/s*] | [*250us*] | | |
| Option 7a | [*DL:9.8Gb/s*]<br>[*UL:15.2Gb/s*] | [*250us*] | | |
| Option 7b | [*DL:9.2Gb/s*]<br>[*UL:60.4 Gb/s*] | [*250us*] | | |
| Option 7c | [*DL:9.8Gb/s*]<br>[*UL:60.4Gb/s*] | [*250us*] | | |
| Option 8 | [*DL:157.3Gb/s*]<br>[*UL: 157.3Gb/s*] | [*250us*] | | |

Note: The values are examples provided by LTE reference, as provided in [11], and are to be replaced by NR values when available.

[1] Description of the split option
[2] Driving feature / use-case requiring a certain split option

# Annex B Change history

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | Meeting | TDoc | CR | Rev | Cat | Subject/Comment | New version |
| 2016-04 | RAN3#91 bis | R3-160947 | | | | Included Editor's notes.<br>Reflected agreed text proposals from RAN3#91bis (R3-160989, R3-161005, R3-161006, R3-161008, R3-161010, R3-161011, R3-161013). | 0.1.0 |
| 2016-05 | RAN3#92 | R3-161381 | | | | Editorial updates. | 0.1.1 |
| 2016-05 | RAN3#92 | R3-161442 | | | | Editorial updates (on referencing).<br>Moved Annex A to section .6.3.1.1 and Annex B to section 6.1.2.1.<br>Reflected agreed text proposals from RAN3#91bis (R3-161009, R3-161012) and RAN3#92 (R3-161296, R3-161344, R3-161444, R3-161449, R3-161453, R3-161485, R3-161486, R3-161495, R3-161503, R3-161504, R3-161506, R3-161507, R3-161508, R3-161509, R3-161510). | 0.2.0 |
| 2016-08 | RAN3#93 | R3-161687 | | | | Added the purpose of this TR in section 1. | 0.3.0 |
| 2016-08 | RAN3#93 | R3-161962 | | | | Reflected agreed text proposals from RAN3#91bis (R3-161867, R3-161972, TP from R3-161973for TR is agreed only the sentence not the new section, R3-161975, R3-162049, R3-161978, R3-161979, R3-161980, R3-162011, R3-161966, R3-162056, R3-162057, R3-162058, R3-162032, R3-162034, R3-162059, R3-162062, R3-162051, R3-162063, R3-162005, R3-161995, R3-161997, R3-161998, R3-162077, R3-162053, R3-162054, R3-162060, R3-162078). | 0.4.0 |